ARTICLE IN PRESS

Economic Modelling xxx (xxxx) xxx

Contents lists available at ScienceDirect

Economic Modelling

journal homepage: www.journals.elsevier.com/economic-modelling

Exploring GDP growth volatility spillovers across countries

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ARTICLE INFO

JEL classification:
F43
G01
C32

Keywords:
GDP growth
Growth volatility
Spillover
Vector autoregression
Variance decomposition

ABSTRACT

This study takes a portfolio approach to investigate GDP growth volatility spillovers among 120 countries of the world during the 1960–2017 period. Based on the ratios of growth rate to standard deviation, we rank these countries in pools of high-, midsize-, and low-income growth. Using a spillover index that is based on variance decompositions under a vector autoregressive framework, we analyze the sources of growth volatility dynamics in the world in terms of volatility proportions from and to others. We find that high-income growth countries are net transmitters, while low-income growth countries are net recipients of growth volatility. We also find that it takes several years for low-income growth countries to absorb growth shocks of global nature. Overall, our analyses illustrate the importance of partnership in risk sharing as it is related to portfolio strategies and risk management.

1. Introduction

Standard deviations of gross domestic product (GDP) growth between 1961 and 2017 for countries classified by the World Bank as high-, midsize-, and low[1]-income were 5.51%, 7.19%, and 10.19%, respectively. Given that the growth variances are much lower than stock market variances[2] and the difference in growth variances is much lower than the potential difference in policy variations among these groups of countries, a robust measure of growth variance is the key for a sound analysis of variance dynamics among heterogeneous countries.

The literature on the determinants of growth is quite extensive (see, Durlauf and Johnson, 2005 for a survey). Growth determinants include business cycle indicators, trade openness, financial flows, financial integration and development, policy volatility, and institutional quality. How these factors may impact growth depends on model sophistications in dealing with issues such as endogeneity, structural breaks, or missing variables. Looking at a sample of findings on these determinants, it can be concluded that growth volatility is negatively related to growth (e.g., Ramey and Ramey, 1995); trade is positively related to growth (e.g., Kose et al., 2006); policy volatility is negatively related to growth (Fatás and Mihov, 2013); institutional quality reduces growth (e.g., Mathonnat et al., 2019); financial development reduces growth (e.g., Easterly et al., 2001); and financial flow is negatively related to growth (e.g., Lensink and Morrisse, 2006; Bugamelli and Paterno, 2011; and Ahamada and Coulibaly, 2011).

While this voluminous literature clearly shows that a country's growth volatility has a component attributable to internal disturbances and another to external disturbances, this important decomposition in volatility is not emphasized enough in the growth literature as volatility dimensions. We divide the sources of growth volatility into domestic and foreign using annual GDP figures of 120 countries under a portfolio framework and the spillover index of Diebold and Yilmaz (2012). Specifically, we investigate the weight components of growth volatility of a given risk-weighted growth volatility portfolio of countries in terms of their own volatility dynamics, volatility impacts on other countries, and received volatility from others, respectively.

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[1] Low income countries comprise primarily the countries of Sub Saharan Africa (excluding mid- and high-income countries), which accounts for most of the less-developed economies in the world.

[2] For instance, between 1973 and 2017, the U.S. stock market showed a standard deviation of 17.01% whereas the Argentina stock market showed a standard deviation of 108%.

https://doi.org/10.1016/j.econmod.2019.11.015
Received 13 March 2019; Received in revised form 11 November 2019; Accepted 12 November 2019
Available online xxxx
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Please cite this article as: Abosedra, S. et al., Exploring GDP growth volatility spillovers across countries, Economic Modelling, https://doi.org/10.1016/j.econmod.2019.11.015

Unlike previous studies using standard deviation estimates on rolling windows,[3] we use risk-weighted standard deviation estimates to examine the question of how countries with substantially different policy variations could be linked in terms of growth volatility. In particular, we focus on linkages between developed and less-developed countries. Surprisingly, very little has been published on growth volatility patterns and its dynamics.

Our main contribution to the growth literature lies in the use of a portfolio approach for summarizing growth volatility risk estimates across countries, which allows the inclusion of virtually all the countries of the world for which GDP figures are available since 1970, while alleviating the endogeneity problem particularly prominent for countries sharing the same information structure.

Previous studies on growth volatility dynamics or growth volatility impacts use either a limited number of countries (e.g., Antonakakis and Badinger, 2016; and Apostolou and Beirne, 2019) or a panel of countries (e.g., Easterly et al., 2001; Lee, 2010; Ahamada and Coulibaly, 2011; and Trypsteen, 2017) to accommodate demanding multivariate autoregressive models in terms of parameter identifications. The portfolio approach is not limited by the number of countries to be analyzed, and yields more robust standard deviation averages for applying to risk-weighted standard deviations across countries, where risk is given by the sensitivity (beta) of a given country's growth relative to some chosen aggregate growth portfolio.

Moreover, our portfolio approach allows different sorting schemes in ranking countries using different macroeconomic characteristics. For instance, we rank countries in terms of the ratios of growth averages to growth residual standard deviation. This gives us an "empirical" rather than a "stylized" ranking and allows us to rank countries in pools of high-, midsize-, and low-income growth, respectively. The fact that our analysis of volatility spillovers involves countries at different development stages suggests that our portfolio approach is potentially superior in terms of devising policies on specific types of countries. Therefore, our approach gives policy makers a better overview of risk exposure of a given country to global disturbances, whereas a panel approach may not provide policy makers with a quick and sharp understanding of a given country's risk profile.

An additional contribution of this study is elucidating the information content of the spillover index of Diebold and Yilmaz (2012), which is actually a matrix embedding three sets of informative entries. It follows from different studies[4] that the spillover index includes vertical entries that represent transmissions to other countries, horizontal entries that represent contributions from other countries, and diagonal entries that represent their own dynamics. For this study, we explore in particular the patterns of the diagonal entries as associated with the order of estimated vector autoregressive (VAR) models. This allows us to examine how quickly volatility shocks are transmitted from high-to low-performing economies.[5] In this respect, we find that it takes several years for low-performing economies to fully absorb volatility shocks from high-performing economies, whereas high-performing economies fully absorb within a year volatility shocks from less-performing economies. As a result, yearly data may not be appropriate for analyzing the volatility dynamics of high-performing economies with more efficient reporting and trading channels. Nonetheless, looking at volatility spillovers of the G7 countries[6] in line with Antonakakis and Badinger (2016), we find that yearly data provide more reliable figures of volatility spillover indices across the G7 countries.[7]

We also examine some of the transmission channels of growth volatility.[8] We highlight three key findings. Trade is an important channel of volatility spillovers among countries, with low-income growth contributing the most to growth volatility. Volatility spillovers attributable to financial depth have declined from about 60% in 1992 to about 25% in 2017 for high-income countries. While broad money and trade show similar volatility shifts, the contribution of foreign direct investment to growth volatility is marginal.

Our findings on volatility dynamics and transmissions underline the importance of partnership in risk sharing as it is related to optimal portfolio strategies, hedging activities, and risk management. While trade balance would show how a country stands vis-à-vis other countries in trade transactions, the analysis of volatility recipients and transmitters yields additional insights into risk partnerships. In fact, our findings on bidirectional spillovers highlight the trade-off existing between transmitting and spreading volatility shocks. For instance, looking at the G7 countries through a VAR (2) model and 5-step-ahead forecasts, the spillover indices show that volatility proportions for European and non-European countries are at 34.05% and 65.05%, respectively. Hence, countries with larger shares of global volatility may be more vulnerable under global economic crises, while countries with smaller shares may be protected from global crisis but may miss out under a booming global economy.

The rest of this study proceeds as follows. Section 2 gives a brief review of spillover approaches and provides an intuitive interpretation of the used volatility spillover index. Section 3 presents the data on GDP figures and opens a discussion on GDP growth patterns. Section 4 reports our findings about volatility dynamics across economies, and Section 5 closes the study with a summary and some concluding remarks.

2. Overview of spillover approaches and insights into the spillover index under consideration

Volatility spillovers are studied in financial economics literature using different approaches. The multivariate generalized autoregressive conditional heteroscedasticity (MGARCH) approach is commonly used to investigate volatility dynamics for highly speculative securities (see Bauwens et al., 2006, for a survey). However, the MGARCH approach requires clustered returns of either sign, which suggests that modelling yearly interlinkages under the MGARCH approach may yield poor results. Econometric approaches constructed using VAR frameworks are more appealing. The Global VAR (GVAR) model of Pesaran and Schuermann (2004) is an example among many others. The GVAR model links economies in terms of countries' macroeconomic variables and trade patterns. Since the GVAR approach requires a full set of macroeconomic variables and trade patterns, the model is over-dimensioned for our purpose.

The Diebold and Yilmaz (2012) spillover index is a valuable tool for analyzing a specific source of volatility interlinkage across countries. It

[3] E.g., Antonakakis and Badinger (2016).

[4] E.g., Diebold and Yilmaz (2009, 2012, 2014), Badinger (2010), Zhou and Zhang (2012); Ben Sita (2013), Awartani and Maghyereh (2013), Cronin (2014), Fengler and Gisler (2015), Antonakakis and Badinger (2016); and Kang et al. (2017).

[5] We use interchangeably the concepts of high-, mid- and less-performing economies (countries) with high-, mid- and low-income economies (countries), respectively.

[6] The G7 countries are Canada (CAN), France (FRA), Germany (GER), Italy (ITA), Japan (JPN), the United Kingdom (UK) and the United States of America (USA).

[7] In Antonakakis and Badinger (2016) volatility spillovers from Canada (CAN) to the United States of America (USA) are 1.7% and 0.9%, and from the USA to CAN are 1.9% and 0.1%, before and after the global financial crisis, respectively. Yearly data on residual standard deviations obtained on a 5-year rolling window yields under a VAR (2) model and a spillover index at 5-year horizon from CAN to the USA is 9.32% and from the USA to CAN is 8.05%. Overall, yearly data gives countries' own volatility dynamics at a lower level, which suggests stronger linkages among countries with yearly than monthly data.

[8] See for instance, Kose et al. (2006); Di Giovanni and Levchenko (2009); Kim et al. (2016); and Haddad et al. (2013) on trade openness, Lensink and Morrisse (2006) on FDI, Ahmad and Coulibaly (2011) on remittances, Apostolou and Beirne (2019) on monetary interventions, and Easterly et al. (2001) on broad money.

uses VAR models and variance decompositions to provide directional spillovers in terms of recipients and transmitters of spillovers. Before delving into the specifics of the spillover index in section (4), let us start by clarifying the intuition shared by a number of studies on financial connectedness (e.g., Diebold and Yilmaz, 2014).

The total variance of a country (TV) can be thought of as a sum of the domestic variance (DV) and the foreign variance (FV). That is, $TV = DV + FV$, where FV represents a variance on a rare event, such as the global financial crisis. By virtue of efficient diversifications, a country diversifies its idiosyncratic effects by efficiently allocating resources across different productive entities, whereas it reduces its global exposure by carefully choosing its bilateral trade partners.

Focusing on the weighting of the domestic and the foreign variance terms, we rewrite the total variance of a country as $TV = W_{ii}^2 DV + W_{ij}^2 FV$, where W_{ii} is the domestic weight and W_{ij} is the foreign weight. Expanding $W_{ij}^2 FV$ into n countries, the total variance of country 1 is,

$$\begin{aligned} TV_{D\to F} &= W_{11}^2 DV + (W_{12} + W_{13} + \ldots + W_{1n})^2 FV \\ TV_{F\to D} &= W_{11}^2 DV + (W_{21} + W_{31} + \ldots + W_{n1})^2 FV \end{aligned} \tag{1}$$

where $TV_{D\to F}$ is what we will refer later to as spillovers "to others," specifically in terms of $(W_{12} + W_{13} + \ldots + W_{1n})$, and $TV_{F\to D}$ as spillovers "from others," essentially in terms of $(W_{21} + W_{31} + \ldots + W_{n1})$. Equation (1) ignores covariance terms, but yields the following intuitions about our research inquiry:

- The weighting scheme represents a country's exposure to foreign volatility risk.
- The higher W_{11} is, the more closed country 1 is with respect to others.
- The spillover index helps estimate $(W_{12} + W_{13} + \ldots + W_{1n}) = W_{TO}$ and $(W_{21} + W_{31} + \ldots + W_{n1}) = W_{FO}$, where W_{TO} is the sum of weights "to others" and W_{FO} is the sum of weights "from others."
- The ratio of W_{FO} to W_{TO} is an indicator for greater risk exposure for $(W_{FO}/W_{TO}) > 1$.
- W_{ij} in $W_{ii}^2 DV + W_{ij}^2 FV$ can be expressed as either W_{FO} or W_{TO}. However, if a country considers both the horizontal and the vertical spillover impacts, W_{ij} must be expressed as a linear combination of W_{FO} and W_{TO}.
- Since $(W_{ii} + W_{ij})$ must be 1, a linear combination of W_{FO} and W_{TO} can be expressed as $W_{ii} + \phi(W_{TO} + W_{FO})$, where $\phi = (1 - W_{ii})/(W_{TO} + W_{FO})$. Hence, ϕ can be viewed as a policy tool. For example, by setting to zero some of the weights in W_{TO} and W_{FO}, a policy maker has a tool for choosing a country's traded partners.

3. Empirical strategies

3.1. Data

We use annual readings of GDP per capita from the World Bank database, which includes GDP figures for 264 economies including purely national economies, sub-continental economies, multinational clubs of economies, and territory economies. We consider two datasets. The first dataset (Sample A) includes 120 economies for which GDP figures are available from 1970 and earlier and is the basis of our analysis of growth volatility dynamics. The second dataset, Sample B, includes 179 economies selected among 264 economies and includes countries with GDP figures available after 1970. We use this sample to compute some measures of macroeconomic volatility. Countries that are neither in Sample A nor in Sample B either do not have GDP figures available or are classified as territories, depending on whether a country belongs to one of the two samples. We filled in the few missing data with previous years' values.

The World Bank Group (WBG) database includes a number of grouping of countries' GDP in terms of their national incomes, geographical location, and economic affinity. For the purpose of examining to which extent our constructed portfolios are correlated with the WBG's grouping in terms of national incomes, our sample of GDP figures include GDP figures for high-income countries (HIC), midsize-income countries (MIC), and the Sub-Saharan Africa (SSA), excluding mid- and high-income countries.

Table 1
Countries/economies ranked by their international symbols.

1. Andorra (AND)	41. Ghana (GHA)*	81. Netherlands (NLD)
2. Argentina (ARG)*	42. Gambia (GMB)*	82. Norway (NOR)*
3. Australia (AUS)	43. Guinea-Bissau (GNB)*	83. Nepal (NPL)
4. Austria (AUT)	44. Greece (GRC)	84. New Zealand (NZL)
5. Burundi (BDI)*	45. Greenland (GRL)	85. Oman (OMN)*
6. Belgium (BEL)	46. Guatemala (GTM)*	86. Pakistan (PAK)*
7. Benin (BEN)*	47. Hong Kong (HKG)	87. Panama (PAN)*
8. Burkina Faso (BFA)*	48. Honduras (HND)*	88. Peru (PER)
9. Bangladesh (BGD)*	49. Indonesia (IDN)*	89. Philippines (PHL)*
10. Bahamas (BHS)*	50. India (IND)*	90. Papua N Guinea (PNG)*
11. Belize (BLZ)*	51. Ireland (IRL)	91. Puerto Rico (PRI)
12. Bermuda (BMU)*	52. Iran (IRN)*	92. Portugal (PRT)
13. Bolivia (BOL)*	53. Iraq (IRQ)*	93. Paraguay (PRY)
14. Brazil (BRA)*	54. Iceland (ISL)	94. Qatar (QAT)*
15. Brunei (BRN)*	55. Israel (ISR)	95. Rwanda (RWA)*
16. Botswana (BWA)	56. Italy (ITA)	96. Saudi Arabia (SAU)*
17. Central Africa R (CAF)*	57. Jamaica (JAM)*	97. Sudan (SDN)*
18. Canada (CAN)	58. Jordan (JOR)*	98. Senegal (SEN)*
19. Switzerland (CHE)*	59. Japan (JPN)	99. Singapore (SGP)
20. Chile (CHL)*	60. Kenya (KEN)*	100. Sierra Leone (SLE)*
21. China (CHN)	61. Cambodia (KHM)*	101. Salvador (SLV)*
22. Cote d'Ivoire (CIV)	62. Kiribati (KIR)*	102. Suriname (SUR)*
23. Cameroon (CMR)*	63. St. Kitts & Nevis (KNA)*	103. Sweden (SWE)*
24. Congo, DR (COD)*	64. Korea, Rep (KOR)*	104. Eswatini (SWZ)
25. Congo, Rep (COG)*	65. Kuwait (KWT)*	105. Seychelles (SYC)*
26. Colombia (COL)	66. Liberia (LBR)*	106. Syria (SYR)*
27. Costa Rica (CRI)*	67. Liechtenstein (LIE)*	107. Chad (TCD)
28. Cuba (CUB)*	68. Lesotho (LSO)	108. Togo (TGO)
29. Germany (DEU)	69. Luxemburg (LUX)	109. Thailand (THA)*
30. Denmark (DNK)	70. Morocco (MAR)	110. Trinidad Tobago (TTO)*
31. Dominican Rep (DOM)*	71. Madagascar (MDG)	111. Tunisia (TUN)
32. Algeria (DZA)	72. Mexico (MEX)*	112. Turkey (TUR)*
33. Ecuador (ECU)*	73. Mali (MLI)	113. Uganda (UGA)*
34. Egypt (EGY)*	74. Malta (MLT)	114. Uruguay (URY)*
35. Spain (ESP)	75. Mauritania (MRT)	115. United States of A (USA)
36. Finland (FIN)	76. Malawi (MWI)*	116. Venezuela (VEN)
37. Fiji (FJI)	77. Malaysia (MYS)*	117. Virgin Islands (VIR)*
38. France (FRA)	78. Niger (NER)*	118. South Africa (ZAF)
39. Gabon (GAB)*	79. Nigeria (NGA)	119. Zambia (ZMB)
40. United Kingdom (GBR)	80. Nicaragua (NIC)*	120. Zimbabwe (ZWE)*

(*) means that the GDP growth rate of country i, Δy_{it}, is not normally distributed.

As we discuss below, we will rank countries/economies in terms of their ratios of growth averages to standard deviations. Though related to the paradigm of risk-return in finance, these ratios may not be robust to the true economic profile of a country. Therefore, for robustness purposes, we will also rank countries in terms of government effectiveness (GEF), regulatory quality (REQ), and rule of law (RUL). Based on the percentile rank given in the WBG database, the lowest rank is 0 and the highest rank is 100. Looking at the long definition[9] of GEF, REQ, and RUL, we understand that these indicators capture deviation from promises and/or commitments. For our purpose, the extent to which a country meets its commitments to trade partners (GEF), to the development of a strong private sector (REQ), and to the enforcement of contract and property rights (RUL) should determine its risk profile.

Similarly, to cope with previous studies on the determinants of

[9] https://datacatalog.worldbank.org/.

growth volatility, we use sample B to obtain measures of macroeconomic volatility. We consider the following macroeconomic variables: total imports (M), total exports (X), broad money (BM), domestic credit to the private sector (CR), and net foreign direct investment (FI).

3.2. GDP growth patterns

Let y_{it} denote the natural logarithm of GDP per capita of country i at time t. We denote change in y_{it} at time t as Δy_{it}, which we interpret as the rate of economic growth. While Chatterjee and Shukayev (2006) argue that the first logarithmic differences of GDP tend to exhibit bias towards a negative relationship between growth and growth volatility, the argument cannot be generalized to volatility analysis. On the other hand, log-differences generally exhibit nice empirical properties including stable averages and standard deviations. Table 1 shows the list of countries ordered by their symbols in the WBG database. Countries with non-normal GDP growth rates are identified with an asterisk (*).

Among the 120 countries in sample A, 46 countries exhibit normal growth processes, which suggest that the averages and the standard deviations of those countries are sufficient to describe their growth processes. In general, advanced economies such as the United States of America (USA) and Japan (JPN) exhibit normal growth processes, whereas normal growth processes are only exceptionally associated with developing countries.

Since the growth rate is not normal for 74 countries, and our aim is to obtain robust mean and standard deviation statistics to regime changes, we test for the presence of breaks or regime changes in growth rates for the 120 countries. In fact, the presence of structural breaks may partially explain why an economic process is not normal.

We use the Bai and Perron (2003) procedure to test for the presence of structural breaks. In this respect, we set the maximum number of possible structural breaks at 5 and the minimum data span of 10 years for a break to occur. Noting that the likelihood of a structural break increases with long spans of data and also that structural breaks induce non-stationarity in the data, 5 structural breaks are more than what 56 years of annual growth rate data can allow. In fact, international evidence points to no break, one break, or two breaks in levels (e.g., Ben-David et al., 2003). Following our findings on the maximum number of breaks across countries, we write the Bai and Perron (2003) model as follows:

$$\Delta y_{it} = \delta_0 + \delta_1 d_{1,i,t} + \delta_2 d_{2,i,t} + \delta_3 d_{3,i,t} + \varepsilon_{i,t}, \tag{2}$$

where Δy_{it} is growth for country i, $d_{b,i,t}$ are indicators for the time of breaks for $b = 1, 2$and 3, δ_b are mean-shifting coefficients, δ_0 is an intercept, and $\varepsilon_{i,t}$ is an error term at time t. Since Δy_{it} is a logarithmic change, the effects of breaks are strongly alleviated as the time of a break may, in the process, be pushed backward or forward on the timeline.

As it stands, equation (2) is based on indicators with start- and end-points, according to Bai and Perron (2003) tests. Table 2 reports a maximum of 3 possible structural break dates for some countries. These dates are illustrative in the sense that at each point in time, the null hypothesis of no structural break may not be rejected.

Looking at Table 2 and referring to equation (2), the break years for Belgium (BEL), for example, occur at 1970, 1980, and after 2007. The indicators in equation (2) are $d_{1,t} = 1$ when $t \leq 1970$ and 0 otherwise, $d_{2,t} = 1$ when $t \leq 1980$ and 0 otherwise, and $d_{3,t} = 1$ when $t \leq 2007$ and 0 otherwise. The Bai and Perron (2003) tests show that 36 among the 120 countries do not exhibit structural breaks in their growth rate processes. However, 25, 42, and 21 countries show 3, 2, and 1 structural breaks, respectively. Since countries show breaks at different years, we also examine the break dates for the world (WLD) GDP growth. The world shows 3 significant breaks with mean-shifting times 1970, 1980, and 2007.

Table 2 shows at least 1 break across countries in all years but 1996. To gain a better overview of the break distribution across countries, we portray the total number of breaks over countries for years between 1970 and 2017. Fig. 1 shows the total number of regime change and mean shift outcomes (breaks) per year in the world.

The total number of breaks ranges from 0 to 54 for an average of

Table 2
Break or mean shifting date points.

AND (1970,1980,2007)	DOM (1970,1980,1990)	KHM (1972,1992,2007)	PRI (1971,1981,2004)
ARG (1972,1992,2002)	DZA (1971,1981,1995)	KIR (1974,1985,2001)	PRT (1974,1985,1995)
AUS (1971,1982,2002)	ECU (1971,1981,2000)	KNA (1970,1990,2007)	PRY (1971,1981,2003)
AUT (1970,1980,1995)	EGY (1974,1985,2004)	KOR (1972,1982,1995)	QAT (1970,1980,1991)
BDI (1972,1982,2003)	ESP (1970,1980,2007)	KWT (1980,1994,2007)	RWA (1971,1981,1994)
BEL (1970,1980,2007)	FIN (1971,1990,2000)	LBR (1975,1985,1995)	SAU (1970,1980,1990)
BEN (1970,1980,1994)	FJI (1970,1980,2001)	LIE (1970,1980,1995)	SDN (1972,1987,1999)
BFA (1970,1980,1994)	FRA (1970,1980,2007)	LSO (1970,1980,2002)	SEN (1970,1980,2001)
BGD (1975,1986,2006)	GAB (1970,1980,1998)	LUX (1974,1985,1995)	SGP (1970,1980,1995)
BHS (1975,1985,1999)	GBR (1970,1980,2007)	MAR (1970,1980,2001)	SLE (1971,1982,2000)
BLZ (1970,1980,2007)	GHA (1980,1991,2001)	MDG (1970,1980,1991)	SLV (1970,1980,1990)
BMU (1970,1981,2007)	GMB (1972,1993,2003)	MEX (1971,1981,2007)	SUR (1982,1992,2007)
BOL (1970,1981,2003)	GNB (1971,1982,1998)	MLI (1970,1980,1994)	SWE (1980,1991,2001)
BRA (1979,1991,2002)	GRC (1979,1989,2007)	MLT (1970,1980,2007)	SWZ (1975,1985,1995)
BRN (1970,1980,1998)	GRL (1970,1980,2007)	MRT (1976,1991,2002)	SYC (1970,1980,1993)
BWA (1970,1980,1990)	GTM (1970,1980,1990)	MWI (1970,1980,1994)	SYR (1983,1994,2007)
CAF (1971,1990,2000)	HKG (1970,1980,1997)	MYS (1970,1980,1996)	TCD (1978,1994,2007)
CAN (1970,1981,2007)	HND (1971,1981,1994)	NER (1970,1980,2000)	TGO (1970,1980,1994)
CHE (1979,1990,2001)	IDN (1971,1981,1998)	NGA (1980,1993,2007)	THA (1972,1995,2005)
CHL (1973,1985,1995)	IND (1970,1980,2002)	NIC (1980,1990,2000)	TTO (1971,1982,1993)
CHN (1978,1993,2007)	IRL (1970,1980,2007)	NLD (1970,1980,2007)	TUN (1970,1980,2007)
CIV (1970,1980,1994)	IRN (1970,1983,1993)	NOR (1970,1980,2007)	TUR (1971,1981,2007)
CMR (1971,1981,2001)	IRQ (1972,1994,2004)	NPL (1971,1981,2003)	UGA (1977,1987,2003)
COD (1978,1989,1999)	ISL (1970,1980,2007)	NZL (1970,1988,2001)	URY (1970,1981,2003)
COG (1982,1994,2007)	ISR (1974,1985,1995)	OMN (1971,1981,1998)	USA (1971,1984,2007)
COL (1971,1981,1991)	ITA (1980,1990,2007)	PAK (1973,1983,2002)	VEN (1970,1981,2003)
CRI (1970,1980,1991)	JAM (1975,1985,1997)	PAN (1972,1982,2003)	VIR (1970, 1997,2007)
CUB (1970,1980,1998)	JOR (1971,1981,1991)	PER (1975,1988,1998)	ZAF (1971,1981,2002)
DEU (1970,1980,1995)	JPN (1978,1995,2006)	PHL (1970,1980,2003)	ZMB (1974,1986,2002)
DNK (1979,1995,2007)	KEN (1970,1980,1993)	PNG (1980,1992,2002)	ZWE (1970,1981,2007)

Note: Structural break/mean shifting date points under Bai and Perron (2003). Break dates for Sub-Saharan African (SSA) countries, mid-income (MIC) countries, high-income (HIC) countries, and the world (WLD) are 1970, 1980, and 2001; 1970, 1980, and 1995; 1970, 1980, and 2007; and 1970, 1980, and 2007; respectively.

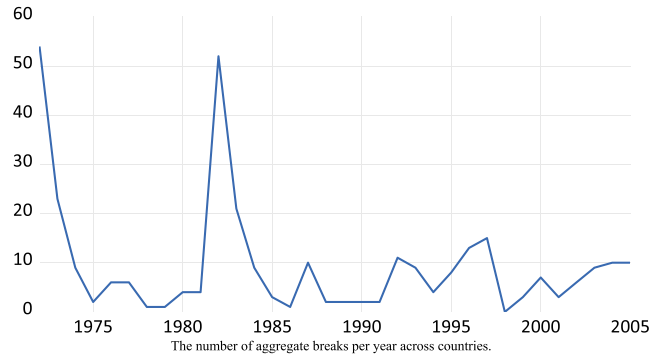


Fig. 1. Break patterns over the sample period.

about 9 breaks per year. The break pattern in Fig. 1 suggests that the world economy has passed through phases of expansions and contractions. This cyclic regularity of phases has been dominated by 3 noticeable peaks of mean shifting events during the last 56 years. A large number of countries in the world were impacted in three years, namely, 1970 with 54 break events, 1980 with 52 break events, and 2007 with 23 break events. The majority of countries showing a break in 2007 turn out to be advanced economies.

The peaks of 1970, 1980, and 2007 are associated with economic events that shook the world economy. The peak of year 1970 can be associated with the 1970s energy crisis and appears as an early warning of the OPEC oil price shock occurring first in 1973. Another oil price shock occurred in 1979 following the Iranian revolution, and several isolated economic actions responsible for many events may explain the peak of year 1980. These include events that led to foreclosures of many saving and loan associations in the USA, bloated public deficits and loose monetary policies in Latin America (Stiglitz, 2000), as well as loan crises in less developed countries (Demirguc-Kunt and Detragiache, 1998). Finally, the peak of 2007 is clearly a reflection of the global financial crisis of 2007/2008.

3.3. Aggregating countries' growth residuals in terms of change per unit risk

Given that the zero-mean error terms $\varepsilon_{i,t}$ in equation (2) are uncorrelated with respective standard deviations σ_i, we obtain estimates of the average growth and standard deviation as $\widehat{\mu}_i = (\delta_0 + \delta_1 + \delta_2 + \delta_3)/\ 4$ and $\widehat{\sigma}_i = \sum_{t=1}^{T-4} \varepsilon_{i,t}^2$, respectively. Table 3 displays the ratio of the average to the standard deviation for each country.

The ratio of $\widehat{\mu}_i$ to $\widehat{\sigma}_i$ in Table 3 represents the number of units of growth per units of risk. We refer to this ratio as the change per unit risk (CUR) since the ratio of $\widehat{\mu}_i$ to $\widehat{\sigma}_i$ represents the reward to a risk point estimate: a higher growth rate should command a higher standard deviation.

Because we can draw on the mean-variance argument, the CUR is more reliable than other measures obtained on primary data or based on deterministic economic activities. Nonetheless, we will, in the course of our analysis, rank countries in terms of REG, REQ, and RUL. Before revealing our ranking in terms of these deterministic parameters based on primary data, Table 3 shows the ranking of the countries under consideration from the highest to the lowest CUR.

The CUR-based ranking displayed in Table 3 randomizes the sample of countries on the top, in the middle, and at the bottom. From a portfolio standpoint, the fact that countries from different regions of the world can make it to the top 40 implies that the ranking achieves a selection that would result in higher diversification benefits. Many studies in the literature rely on factual rankings such as G7 or OECD countries, which provide a myopic view of the marginal gain/loss achieved every year by any country. A ranking that would group high-performing countries together does not give a true picture of the marginal change in growth across countries. Thus, the CUR-based ranking shows how global shocks would travel among economically and/or geographically distant countries. For instance, the first 40 countries include the USA (North America), Panama (Central America), Bolivia (Latin America), Norway (Europe), Botswana (Africa), Fiji (Oceania), and China (Asia), which means that the 40 first economies represent the world in a nutshell. Hence, the path of a global shock born in a given country can be tracked through this group of countries, though imperfectly. In contrast, the majority of countries making the last 40 are from SSA, which could be an indication that a global shock from one of the top countries may affect this group with longer lags, and that the volatility dynamics in this group may diverge from other regions.

Therefore, we follow, for instance, Mallick and Moore (2008), and divide the 120 countries into three groups, with the first 40 countries falling into the group of countries with a high-income growth, the next 40 into the group of countries with a midsize-income growth, and the last 40 into the group of low-income growth. We refer to the first, the second, and the third group as "high," "mid," and "low," respectively. The pooling of countries requires some understanding of what could be a country's contribution to global risk. Using the average sensitivity of a country's growth residual to HIC, MIC, and SSA growth residuals, we obtain the weight of a country in group p as,

Table 3
Countries ranked by the average to the residual standard deviation (CUR) ratio.

	$\widehat{\mu}_i/\widehat{\sigma}_i$		$\widehat{\mu}_i/\widehat{\sigma}_i$		$\widehat{\mu}_i/\widehat{\sigma}_i$
1. PRI	3.724	41. GTM	0.683	81. BFA	0.392
2. USA	3.113	42. PRY	0.660	82. JOR	0.389
3. HKG	1.638	43. ISR	0.659	83. CHL	0.382
4. PAN	1.620	44. ITA	0.643	84. BRA	0.378
5. BMU	1.429	45. DNK	0.637	85. ZWE	0.377
6. SGP	1.278	46. DEU	0.634	86. JAM	0.370
7. BWA	1.062	47. PAK	0.624	87. BEN	0.369
8. SLV	1.061	48. OMN	0.587	88. TGO	0.365
9. CHN	1.057	49. LSO	0.581	89. QAT	0.349
10. SYC	1.056	50. GRC	0.580	90. MRT	0.348
11. IRL	0.999	51. CMR	0.573	91. PNG	0.347
12. AUT	0.960	52. NPL	0.568	92. BGD	0.345
13. NOR	0.928	53. FIN	0.567	93. CHE	0.333
14. PRT	0.913	54. ISL	0.558	94. IRN	0.302
15. MLT	0.908	55. ECU	0.558	95. SEN	0.297
16. KNA	0.894	56. IDN	0.546	96. RWA	0.295
17. HND	0.877	57. COL	0.543	97. TUR	0.292
18. NLD	0.866	58. EGY	0.540	98. SDN	0.291
19. CAN	0.856	59. ZAF	0.527	99. NER	0.272
20. TUN	0.851	60. CRI	0.526	100. GNB	0.264
21. BLZ	0.845	61. CIV	0.526	101. SYR	0.243
22. ESP	0.811	62. KEN	0.506	102. TCD	0.243
23. VIR	0.792	63. PER	0.500	103. KHM	0.240
24. LIE	0.777	64. SWE	0.498	104. UGA	0.236
25. LUX	0.776	65. DZA	0.492	105. MWI	0.230
26. IND	0.767	66. SWZ	0.486	106. NGA	0.206
27. BEL	0.761	67. BRN	0.483	107. GHA	0.205
28. FRA	0.759	68. TTO	0.482	108. MDG	0.202
29. MAR	0.751	69. MLI	0.480	109. COG	0.194
30. THA	0.744	70. MEX	0.459	110. SUR	0.191
31. BOL	0.736	71. GAB	0.457	111. ZMB	0.176
32. AUS	0.731	72. NIC	0.455	112. CAF	0.158
33. MYS	0.728	73. PHL	0.452	113. SLE	0.157
34. GBR	0.726	74. SAU	0.423	114. KIR	0.141
35. KOR	0.725	75. AND	0.419	115. ARG	0.130
36. FJI	0.725	76. URY	0.418	116. KWT	0.110
37. BHS	0.717	77. VEN	0.410	117. GMB	0.082
38. GRL	0.717	78. DOM	0.404	118. IRQ	0.072
39. JPN	0.702	79. NZL	0.396	119. COD	−0.027
40. CUB	0.686	80. BDI	0.393	120. LBR	−0.057

$\Delta y_{it} = \delta_1 d_{1,i,t} + \delta_2 d_{2,i,t} + \delta_3 d_{3,i,t} + \delta_4 d_{4,i,t} + \varepsilon_{i,t}, d_{k,i,t}$ are break dates (Table 3), $\widehat{\mu}_i = (\delta_1 + \delta_2 + \delta_3 + \delta_4)/4$, and $\widehat{\sigma}_i$ is the standard deviation of $\varepsilon_{i,t}$.
Change per unit risk is the ratio of $\widehat{\mu}_i$ to $\widehat{\sigma}_i$.

Table 4
Beta-sensitivity of country's growth residual to income-based GDP growth residual.

	$\widehat{\beta}_{i,ssa}$	$\widehat{\beta}_{i,mic}$	$\widehat{\beta}_{i,hic}$		$\widehat{\beta}_{i,ssa}$	$\widehat{\beta}_{i,mic}$	$\widehat{\beta}_{i,hic}$		$\widehat{\beta}_{i,ssa}$	$\widehat{\beta}_{i,mic}$	$\widehat{\beta}_{i,hic}$
AND	−0.005	−0.013	1.935*	GHA	−0.020	0.684	0.731	NLD	−0.242*	0.032	1.987*
ARG	−0.170	0.841	1.138	GMB	−0.084	0.839*	−0.153	NOR	0.111	0.367*	1.075*
AUS	0.376*	0.818*	−0.059	GNB	0.232	0.422	−0.161	NPL	0.210	0.331	−0.101
AUT	−0.127	−0.052	1.976*	GRC	−0.065	0.000	1.461*	NZL	0.024	0.250	1.233*
BDI	−0.023	0.437*	0.137	GRL	−0.075	−0.326	1.976*	OMN	0.430	1.760*	−1.386
BEL	−0.221*	−0.024	2.129*	GTM	0.159	0.422*	−0.777*	PAK	0.159	0.295	−0.281
BEN	0.066	0.026	1.279*	HKG	0.053	0.173	0.275	PAN	−0.024	0.113	−0.046
BFA	−0.128	0.082	1.622*	HND	0.149	−0.054	−0.100	PER	0.608*	0.460	−0.590
BGD	0.309	0.208	−0.060	IDN	0.348*	1.075*	−0.564	PHL	0.103	0.524*	0.048
BHS	0.081	0.050	−0.156	IND	0.087	0.253	0.273	PNG	0.141	0.652*	0.435
BLZ	0.173	0.102	0.313	IRL	−0.235	−0.247	1.725*	PRI	−0.012	0.005	0.093
BMU	0.037	0.089	−0.079	IRN	−0.290	1.090	0.893	PRT	0.045	−0.147	1.554*
BOL	0.232	0.578*	−0.864	IRQ	−0.441	2.046*	−0.610	PRY	0.301	0.601*	0.574*
BRA	0.374	1.509	−0.442	ISL	0.029	−0.215	1.879*	QAT	0.881*	1.158*	−0.615
BRN	1.078*	1.176*	−1.079	ISR	0.468*	0.033	0.255	RWA	0.308	−0.043	0.229
BWA	0.416*	−0.226	1.449*	ITA	−0.027	−0.155	1.797*	SAU	0.759	1.103	−0.687
CAF	−0.201	0.033	1.298*	JAM	0.370*	0.187	−0.554	SDN	0.066	0.844	−0.413
CAN	0.283*	0.302*	0.194	JOR	−0.027	−0.173	0.766	SEN	0.180	−0.321	1.551*
CHE	0.776	−0.123	0.088	JPN	0.066	−0.488*	1.574*	SGP	0.447*	0.486*	−0.373
CHL	0.575*	0.139	0.331	KEN	0.016	0.076	0.895*	SLE	0.214	0.688	−0.544
CHN	−0.304*	1.151*	−0.201	KHM	−0.804*	1.227	−0.242	SLV	−0.075	0.302*	0.014
CIV	−0.081	−0.236	1.532*	KIR	0.555	0.417	−0.046	SUR	0.158	0.070	0.500
CMR	0.030	−0.075	0.914*	KNA	−0.229	0.390*	0.179	SWE	0.120	0.108	1.600*
COD	1.751*	0.012	−1.408	KOR	0.218	0.415	0.617	SWZ	0.755*	−0.301	0.799
COG	0.507	0.454	0.813	KWT	0.780	1.169*	−0.341	SYC	−0.063	−0.021	1.065*
COL	−0.195	1.123*	0.232	LBR	0.629	−0.126	−0.608	SYR	−0.082	0.687*	−0.672
CRI	−0.371	0.417	0.766	LIE	−0.113	−0.151	1.986*	TCD	−0.116	0.264	1.493*
CUB	−0.262	0.213	0.578*	LSO	0.833*	−0.452	1.105*	TGO	−0.024	−0.013	1.748*
DEU	−0.238*	−0.041	2.174*	LUX	−0.098	0.061	1.927*	THA	0.180	0.292	0.631*
DNK	−0.192*	−0.056	1.988*	MAR	0.006	0.175	1.225*	TTO	0.176	1.040*	−0.165
DOM	0.327	0.282	−0.623	MDG	−0.435	0.723*	0.831*	TUN	0.045	0.125	0.724*
DZA	0.041	0.938*	0.639*	MEX	−0.056	1.060*	−0.648	TUR	−0.156	0.842	1.161*
ECU	0.038	1.293*	−0.981*	MLI	−0.255	−0.114	1.763*	UGA	0.298	0.309	−0.541
EGY	0.164	0.400	−0.660*	MLT	0.060	−0.190	1.316*	URY	0.205	0.753	0.684
ESP	−0.074	−0.113	2.019*	MRT	0.105	0.529*	0.608*	USA	−0.041	0.023	0.157*
FIN	0.094	0.346	1.227*	MWI	0.081	0.866	−0.563	VEN	0.585	0.253	−0.581
FJI	−0.020	0.417	0.799*	MYS	0.282*	0.751*	0.051	VIR	−0.279	0.082	0.325
FRA	−0.208*	−0.093	2.017*	NER	0.256	−0.442*	1.799*	ZAF	1.010*	−0.380	0.532*
GAB	0.059	1.523*	0.469	NGA	1.912*	0.230	−0.835	ZMB	0.742*	0.685	−0.321
GBR	0.116	−0.141	1.271*	NIC	−0.417	0.439	0.501	ZWE	0.478*	0.028	−0.730

$\varepsilon_{i,t} = \beta_{i,ssa}\varepsilon_{ssa,t} + \beta_{i,mic}\varepsilon_{mic,t} + \beta_{i,hic}\varepsilon_{hic,t} + v_t$; $\varepsilon_{ssa,t}$, $\varepsilon_{mic,t}$, and $\varepsilon_{hic,t}$ are residuals from equation (1). GDP figures of SSA = Sub-Saharan Africa, MIC = mid-income countries, and HIC = high-income countries, respectively. (*) means the estimate is significant at 5% level.

$$w_{i,p} = \overline{\beta}_{i,p} \Big/ \sum_{i=1}^{40} \overline{\beta}_{i,p}\overline{\beta}_{i,p} = (\beta_{i,ssa} + \beta_{i,mic} + \beta_{i,hic})/3\varepsilon_{i,t}$$
$$= \beta_{i,ssa}\varepsilon_{ssa,t} + \beta_{i,mic}\,\varepsilon_{mic,t} + \beta_{i,hic}\varepsilon_{hic,t} + v_t \tag{3}$$

where $p = low,..,high$; v_t is a white noise error term, whereas $\beta_{i,ssa}$, $\beta_{i,mic}$, and $\beta_{i,hic}$ are the sensitivities (beta) of a country with respect to $\varepsilon_{ssa,t}$, $\varepsilon_{mic,t}$, and $\varepsilon_{hic,t}$ from equation (2), respectively. Table 4 reports the coefficient estimates of equation (3).

Table 4 shows that countries have more often the right sign on significant beta with respect to their assigned groups. Some exceptions do appear, however, with few African countries showing positive sensitivity to HIC, and Israel exhibiting a positive beta on SSA growth residuals. Surprisingly, Sub-Saharan African countries are not strongly related to SSA, with only 7 countries showing a positive significant beta among the 26 countries with a significant estimate. High-income countries from Europe are, along with China, negatively relatively related to SSA growth residuals, which may suggest that European countries and China gain from their countercyclical investments in Sub-Saharan Africa. The range of $\widehat{\beta}_{i,ssa}$ is from −0.804 to 1.912 for an average of 0.148.

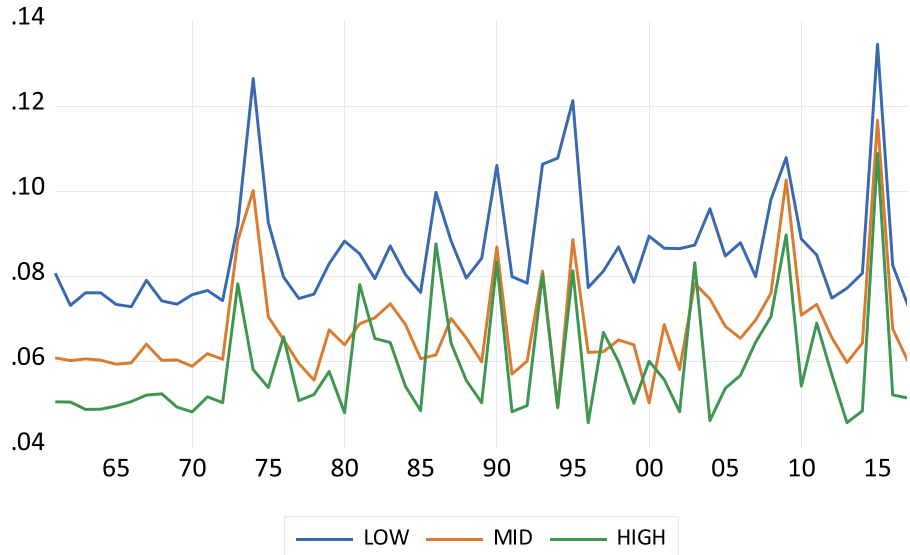
Sensitivities to MIC residuals are limited to 32 countries; all showing a positive estimate with the exception of Japan (JPN) and Niger (NER). In general, countries with significant positive MIC beta are classified as mid-income countries and are virtually from all the continents. The range of $\widehat{\beta}_{i,mic}$ is from −0.488 to 2.046 for an average of 0.339.

In comparison to the SSA and MIC groups, the HIC group is more informative and provides a clearer picture of countries known as high-income countries. Among the 52 countries showing significant sensitivities to the HIC factor, those with negative estimates are known to be mid-income countries (e.g., EGY, ECU, and GTM). The parameters $\widehat{\beta}_{i,hic}$ range from −1.408 to 2.174 for an average value of 0.492. On the basis of average SSA, MIC, and HIC estimates, we can say that sensitivities to SSA, MIC, and HIC increase with income.

Looking at the second expression of equation (3), the subscript p in $\overline{\beta}_{i,p}$ is assigned to portfolio p in terms of CUR, whereas $\overline{\beta}_i$ is the country's average beta. The range of $\overline{\beta}_i$ is from −0.074 to 0.684 for an average of 0.326. Hence, the beta-weighted growth residual at year t are,

$$\varepsilon_{p,t} = \sum_{i=1}^{40} w_{i,p}\varepsilon_{i,p} \tag{4}$$

The correlation between $\varepsilon_{high,t}$ and $\varepsilon_{hic,t}$, $\varepsilon_{mid,t}$, between $\varepsilon_{hic,t}$, and $\varepsilon_{low,t}$, and between $\varepsilon_{ssa,t}$ are 0.952, 0.735, and 0.769, respectively. Moreover, regressing $\varepsilon_{p,t}$ on a constant, $\varepsilon_{ssa,t}$, $\varepsilon_{mic,t}$, and $\varepsilon_{hic,t}$, the adjusted coefficients of determination associated with $\varepsilon_{low,t}$, $\varepsilon_{mid,t}$, and $\varepsilon_{high,t}$ are 0.748, 0.847, and 0.905, respectively. It follows that the constructed portfolios are strongly related to income-based portfolios, which suggests that the CUR scheme does rank countries with some good level of accuracy. The question of how low-income countries are related, with respect to growth



Volatility estimates are based on residuals of individual countries that are grouped into three portfolios in terms of residual variances. LOW, MID, and HIGH are low-, mid-, and high-income growth group, respectively.

Fig. 2. Growth residual volatility patterns in the world between 1961 and 2017.

residual shocks, to high-income countries can now be investigated using $\varepsilon_{low,t}$, $\varepsilon_{mid,t}$, and $\varepsilon_{high,t}$ within a VAR model of appropriate order.

However, the growth residuals may not be very informative in capturing lead-lag dynamics among countries that are geographically or/and economically distant. Therefore, in line with Diebold and Yilmaz (2012), we use volatility estimates in examining volatility interdependence among countries. From the growth residuals of equation (2) and the weighing schemes of equation (3), the portfolio theory suggests the following variance estimate at time t,

$$\delta_{p,t}^2 = w_{i,p}^{'}\left(D_{t,p}\rho_p^{'}D_{t,p}^{'}\right)w_{i,p} + w_{i,p}^{'}\,\Sigma_p w_{i,p}, \tag{5}$$

where $\delta_{p,t}^2$ is the portfolio variance for portfolio p; $D_{t,p}$ is a diagonal matrix of residuals of portfolio p at time t; $w_{i,p}$ are weights of countries in portfolio p, ρ_p is the correlation matrix of residuals of equation (2) over the sample period, and Σ_p is the covariance matrix of residuals over the sample period. The first part of equation (5) is stochastic, as it depends on time t, whereas the second part of equation (5) is positive and constant over the sample period. Equation (5) is consistent with model of volatilities with a constant correlation but a time-varying variance (e.g., Bollerslev, 1990). Fig. 2 shows the volatility patterns of the high-, mid-, and low-income growth countries, respectively.

Fig. 2 shows that volatility across the three groups of countries exhibits peaks in 1972, 1995, 2008, and 2016. These peaks do not exactly match the peaks of Fig. 1 portraying the aggregate patterns of breaks across the countries of the world.

The volatility ranges from 7.3% to 13.5% for the low-income countries, with an average value of 8.6% per year, from 4.9% to 11.7% for mid-income countries with an average value of 6.8% per year, and from 4.6% to 10.9% for high-income countries with an average value of 5.9% per year.

Fig. 2 also shows that growth residual volatility reverts to some sort of long-run averages. Looking at the patterns of the three growth portfolio volatilities, less-performing economies are the most volatile, while high-performing economies are the least volatile. Moreover, Fig. 2 shows that the growth residual volatility curve of midsize economies lies, in general, between the growth residual volatility curves of less- and high-performing economies.

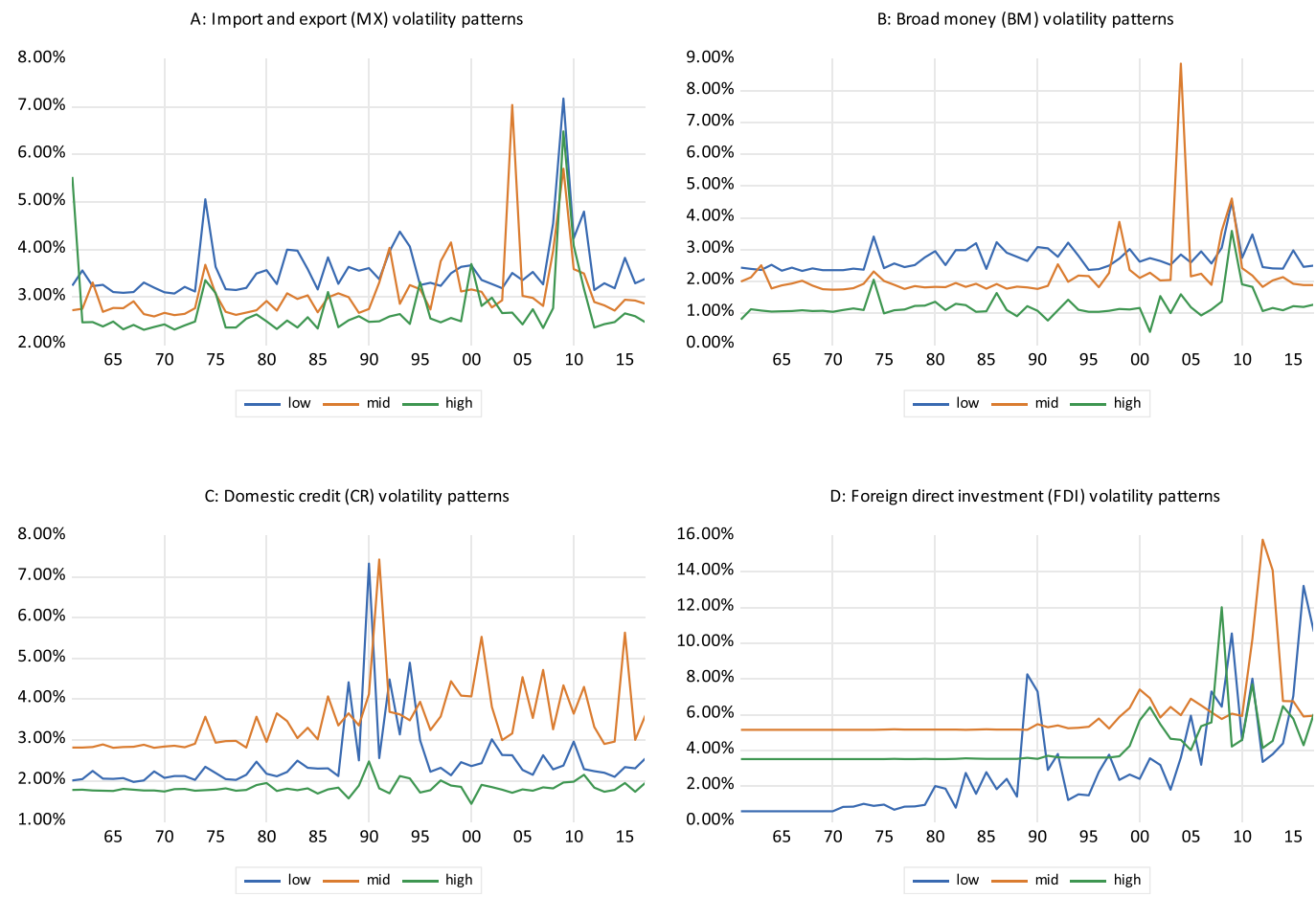
Despite the fact that the three volatility patterns differ in magnitude, a clear co-movement pattern emerges over time indicating that countries seem to rise and fall together. However, only a closer analysis can tell whether volatility increases or decreases in tandem for low-, mid-, and high-income growth countries.

3.4. Determinants of growth volatility

Growth volatility is part of the so-called macroeconomic volatility, which involves systematic effects attributable to economic activities, frictions, and imperfections. The nature of these effects suggests that the sources of macroeconomic volatility are multidimensional in nature. A combination of stock market and fundamental volatility gives, in theory, the best perspective on systematic volatility. However, because macroeconomic volatility intensity co-moves with business cycles rather than with speculative market volatility, we select determinants of growth volatility in the isolation of stock market volatility.

There is virtually an unlimited number of macroeconomic effects that can be related to growth volatility. In this study, we concentrate on the proposition that economic growth is positively related to financial sector development since the tradeoff between consuming today and tomorrow is smoother with the existence of an efficient capital market. Reflecting on this theory, Easterly et al. (2001) found that the more developed the financial sector of a country is, the lower is the volatility in growth (output).

Several measures of financial development have been proposed in the literature. In this study, we use the ratio of foreign direct investment (FDI) to GDP, which, according to Lensink and Morrisse (2006), relates negatively to economic growth volatility. Alternative measures of financial development include the CR to GDP ratio, and BM to GDP ratio (Easterly et al., 2001). Trade openness is another channel of know-how and technology transmission across countries that has been investigated in the literature on volatility interdependence. Indeed, according to the compensation hypothesis, trade openness may lead to excess growth volatility. In this context, Razin et al. (2003) found that trade openness may adversely affect growth by exacerbating uncertainties in macro fundamentals, and Malik and Temple (2009) found that countries that are far from world's economic centers and have weak institutional arrangements experience greater volatility in output growth. In this respect, diversification benefits would be short of international gains in terms of economic development and transfer of technology (e.g., Mireku et al., 2017; and Huang et al., 2014). As a proxy to trade openness, we use



Note: MX, BM, CR, and FI stand for import plus export, broad money, domestic credit to the private sector, and foreign direct investment residual volatility, respectively. The vertical axis gives standard deviation ranges.

Fig. 3. Volatility patterns of potential determinants of GDP growth volatility.

the ratio of the sum of total imports and exports to GDP (MX) following Easterly et al. (2001).

Since BM, CR, FDI, and MX are given as proportions of GDP, we add one to these ratios to get a gross ratio on the basis of which a first logarithmic difference can be computed. We use sample B to get countries' residuals from equation (1). We use the WBG's world aggregate for each macroeconomic variable to obtain the sensitivity of a country's macro effect to the world effect. Therefore, $\overline{\beta}_{i,p}$ is not an average, but $\beta_{i,p}$ is. We sort the sample of 179 countries by CUR and divide them into low-income growth (60 countries), mid-income growth (59 countries), and high-income growth (60 countries). The combination of countries within each group is the beta-weighted sum of variance-covariance in line with equation (4). Fig. 3 portrays the volatility patterns of MX, BM, CR, and FDI for low-, mid- and high-income growth countries, respectively.

The 4 exhibits of Fig. 3 show that macroeconomic volatility is much lower than stock market volatility, even for the most performing economy in the world, the United States of America showing a standard deviation of 17.01% between 1970 and 2017 on yearly stock market data. Macroeconomic volatility ranges between 0.40% on BM volatility and 15.80% on FDI volatility. Despite these relatively low volatility figures, macroeconomic volatility has grown dynamically over the years, with the three groups of countries exhibiting similarities in volatility co-movements.

Looking at each particular macroeconomic volatility pattern, countries show over the years more similarities in MX volatility movements. Thus, imports and exports may be an important channel of volatility spillovers among countries. Volatility patterns on BM across groups differ markedly. Low-income growth volatility curve is, in general, above mid- and high-income growth volatility curves. Mid- and high-income growth volatility curves show more variation in the 2000s. We suspect the use of bailouts in the 2000s to deal with market imperfections during the time of financial crisis as being a factor explaining higher variation in BM volatility of the mid- and high-income countries.

Mid-income growth countries show in general higher variation in CR. Exceptionally, low-income growth countries show higher variation in CR between 1985 and 1995. Bank crises as discussed in Demirguc-Kunt and Detragiache (1998) may explain why low-income growth countries only exhibit higher volatility in the mid-1980s and mid-1990s as opposed to high-income growth countries exhibiting healthy time-varying volatility processes on domestic credit. While the CR volatility pattern does not define a typical low-income growth country's volatility pattern, the FDI volatility pattern does, and noticeably so. The sharp contrast in FDI volatility patterns between the low- and high-income growth countries suggests that their sources of financing virtually differ. For low-income growth countries, FDI has been an important source of volatility, while for high-income growth countries, the variation in FDI may have possibly contributed to their overall volatility only in the late 2000s.

4. Empirical results

4.1. The econometrics of volatility interdependence

Section 3 of our analysis of GDP growth and residuals of the four macroeconomic variables gives 15 volatility portfolios. While our focus is

on GDP residual volatility dynamics, we extend our analysis on channels of growth volatility dynamics by looking at models including growth volatility portfolios and each of the four macroeconomic volatility portfolios.

For our econometric models, we take advantage of the Diebold and Yilmaz (2012) exposition of what they refer to as the volatility spillover index. Here, we limit ourselves to the spillover index equation, referring the reader back to Diebold and Yilmaz (2012) for a more elaborate econometric exposition of the index. As such, the spillover index is based on a structural VAR model and the dynamics of its resulting impulse responses as,

$$\theta_{pq}(H) = \frac{\sigma_{qq}^{-1}\sum_{h=0}^{H-1}\left(e_p^{'}\ \Psi_p \Sigma e_q^{'}\right)^2}{\sum_{h=0}^{H-1} e_p^{'}\ \Psi_p\ \Sigma\ \Psi_p^{'}\ e_p^{'}}, H = 1, 2, \ldots, \tag{6}$$

where σ_{qq} is the standard deviation of the *qth* equation in the VAR equation, Ψ_p are impulse response coefficients, e_p is an indicator taking 1 when group p is selected, and 0 otherwise, Σ is the variance-covariance matrix from the VAR model, and H is the forecasting horizon.

Clearly, $\theta_{pq}(H)$ is a variance-covariance matrix with a diagonal giving the proportion of the total variance attributable to group p while the off-diagonal entries include horizontal and vertical impacts. The horizontal impact is the sum of row-wise entries except the *pp*-entry, whereas the vertical impact is the sum of column-wise entries except the*pp*-entry.

The overall variance proportion is the ratio of the sum of row-wise entries excluding all *pp*-entries to the sum of column-wise entries including all *pp*-entries. The higher this ratio, the stronger is the co-movement among countries.

It is worth noting that the covariance-matrix is not necessarily symmetric: the entry a_{12} in row 1 and column 2 may differ from the entry a_{21} in row 2 and column 1. Since entry a_{12} is the contribution of country 2 to country 1's variance, and entry a_{21} is the contribution of country 1 to country 2's variance, we can think of the net spillover performance as the difference between "to others" and "from others" over "from others" or $\phi_{12} = (a_{21} - a_{12})/a_{12}$.

4.2. Growth volatility dynamics across G7 countries

Antonakakis and Badinger (2016) examined volatility linkages among the G7 countries and found that the USA has been the largest transmitter of volatility shocks to others. Since our analysis of volatility dynamics is based on constructed portfolios, we start our empirical report by examining the G7 countries that are all part of high-income countries. Table 5 reports estimates of indices for the G7 countries exhibiting a total spillover index of 52.70% that is higher than in Antonakakis and Badinger (2016), where the spillover index is estimated on monthly output growth and volatility.

The diagonal entries in bold represent own contribution to volatility dynamics. The higher is the diagonal entry, the lower are the spillovers to others and from others. Table 5 shows a clear contrast between European countries with relatively lower diagonal entries, and non-European countries with a higher degree of self-reliance in volatility dynamics. The reason for this discrepancy is that the four European countries are members of the European Union. Looking at the horizontal impact (W_{FO}) and vertical impact "to others" (W_{TO}), we find that volatility interlinkages are stronger among European countries.

Table 5 also shows that the adjusting factor, ϕ, is nonlinear in diagonal entries. Plotting the diagonal entries against the adjusting factor reveals that the curve bends at $\phi = 0.598$. It follows that the adjusting factor is an intuitive way to linearly combine the vertical and the

Table 6
Volatility decomposition and dynamics over time.

Panel A: Dynamic volatility decomposition based on a VAR model of order 1

	low	*mid*	*high*	From others
low	48.21	27.70	24.09	51.80
mid	27.85	44.52	27.63	55.50
high	23.13	30.04	46.83	53.20
Contribution to others	51.00	57.70	51.70	160.40
Contribution plus own	99.20	102.30	98.50	53.50%

Panel B: Dynamic volatility decomposition based on a VAR model of order 12

	low	*mid*	*high*	From others
low	30.12	22.65	47.22	69.90
mid	23.31	26.45	50.24	73.50
high	25.39	28.02	46.59	53.40
Contribution to others	48.70	50.70	97.50	196.80
Contribution to plus own	78.80	77.10	144.10	65.60%

Panel C: Own volatility dynamics over time

#lags	$\omega_{low,low}$	$\omega_{mid,mid}$	$\omega_{high,high}$	$\Delta\omega_{low,low}$	$\Delta\omega_{mid,mid}$	$\Delta\omega_{high,high}$
1	0.4821	0.4452	0.4683			
2	0.4829	0.4389	0.4512	0.0008	−0.0063	−0.0171
3	0.4768	0.4377	0.4524	−0.0061	−0.0012	0.0012
4	0.4699	0.437	0.4471	−0.0069	−0.0007	−0.0053
5	0.4507	0.4223	0.4445	−0.0192	−0.0147	−0.0026
6	0.4421	0.4248	0.4455	−0.0086*	0.0025	0.0010
7	0.4734	0.4278	0.4482	0.0313	0.0030	0.0027
8	0.4246	0.4122	0.4359	−0.0488	−0.0156	−0.0123
9	0.4275	0.3977	0.4766	0.0029	−0.0145	0.0407
10	0.4161	0.3624	0.5098	−0.0114	−0.0353	0.0332
11	0.3579	0.3074	0.4224	−0.1240	−0.0784	0.0762
12	0.3012	0.2645	0.4659	−0.0590	−0.0038*	−0.0990

p = *low*, *mid and high*, *respectively*; $\omega_{p,p}$ is the proportion of own variance to total variance as a function of impulse responses at horizon 5; and $\Delta\omega_{p,p}$ is change in $\omega_{p,p}$. (*) means that the average is significant at 5% level.

Table 5
Volatility spillover indices on the G7 countries.

	W_{CAN}	W_{FRA}	W_{GER}	W_{ITA}	W_{JPN}	W_{UK}	W_{USA}	W_{FO}
W_{CAN}	0.6334	0.0592	0.1122	0.0228	0.0245	0.0675	0.0805	0.3667
W_{FRA}	0.0319	0.3225	0.2900	0.1518	0.1540	0.0435	0.0063	0.6775
W_{GER}	0.0488	0.2848	0.3205	0.1368	0.1508	0.0461	0.0121	0.6794
W_{ITA}	0.0217	0.2587	0.2259	0.3154	0.0497	0.1052	0.0234	0.6846
W_{JPN}	0.0231	0.1096	0.1085	0.0236	0.6898	0.0442	0.0013	0.3103
W_{UK}	0.0549	0.1381	0.0961	0.1823	0.0124	0.4035	0.1127	0.5965
W_{USA}	0.0932	0.0625	0.0186	0.0167	0.0865	0.0942	0.6283	0.3531
W_{TO}	0.2740	0.9130	0.8510	0.5340	0.4780	0.4010	0.2360	3.6870
$W_{ii} + W_{TO}$	0.9070	1.2350	1.1720	0.8490	1.1680	0.8040	0.8650	0.5270
ϕ	0.5722	0.4260	0.4440	0.5618	0.3935	0.5980	0.6310	

Notes: The indices are on a VAR model of order 2, and variance decompositions on 10-step-ahead forecasts. W_{FO} is the horizontal sum of weights excluding the diagonal entry and measures the impacts of other countries on country ii. W_{TO} is the vertical sum of weights excluding the diagonal entry and measures the impact of country ii on other countries. $W_{ii} + W_{TO}$ is the contribution of others including own. $\phi = (1 - W_{ii})/(W_{TO} + W_{FO})$ is an adjusting factor so that.$W_{ii} + \phi(W_{TO} + W_{FO}) = 1$.

Table 7
Pairwise net volatility spillover dynamics.

#Lags	$\phi_{low,mid}$	$\phi_{low,high}$	$\phi_{mid,low}$	$\phi_{mid,high}$	$\phi_{high,low}$	$\phi_{high,mid}$
1	0.0054	−0.0399	−0.0054	0.0872	0.0415	−0.0802
2	0.0397	−0.0376	−0.0382	0.1069	0.0391	−0.0966
3	0.0178	−0.0423	−0.0175	0.0959	0.0441	−0.0875
4	−0.0123	−0.0090	0.0125	0.0983	0.0090	−0.0895
5	−0.0840	−0.0310	0.0917	0.0837	0.0320	−0.0773
6	−0.1288	−0.0109	0.1479	0.0703	0.0110	−0.0657
7	−0.0544	−0.0933	0.0576	0.1583	0.1028	−0.1367
8	−0.0142	−0.1170	0.0144	0.1172	0.1325	−0.1049
9	0.1762	−0.2477	−0.1498	−0.0706	0.3292	0.0759
10	0.2041	−0.3403	−0.1695	−0.3213	0.5159	0.4733
11	0.4186	−0.3271	−0.2951	−0.2909	0.4862	0.4102
12	0.0291	−0.4623	−0.0283	−0.4423	0.8598	0.7930
$\mu_\phi(half)$	−0.0270	−0.0284*	0.0318	0.0904*	0.0295*	−0.0828*
$\sigma_\phi(half)$	0.0267	0.0061	0.0295	0.0052	0.0064	0.0044
$\mu_\phi(all)$	0.04978	−0.14652*	−0.03165	−0.02560	0.21693*	0.08452
$\sigma_\phi(all)$	0.04334	0.04511	0.03512	0.05962	0.07882	0.08759

Note: net volatility spillover performance = $\phi_{p,q} = (a_{rc} - a_{cr})/a_{cr}$, where a_{rc} = the entry in the matrix correponding to a group's contribution to others, = the entry in the matrix correponding to the contribution from others to the group, $\mu_\phi(half)$ and $\mu_\phi(all)$ are averages over 6 and 12 lags respectively; $\sigma_\phi(half)$ and $\sigma_\phi(all)$ are standard errors of the averages over 6 and 12 lags, respectively; and (*) means significance at 5% level.

horizontal impacts. Since the weights sum up to 100% in this fashion, the estimation of spillover indices could be used as a first step in obtaining weights in a portfolio analysis.

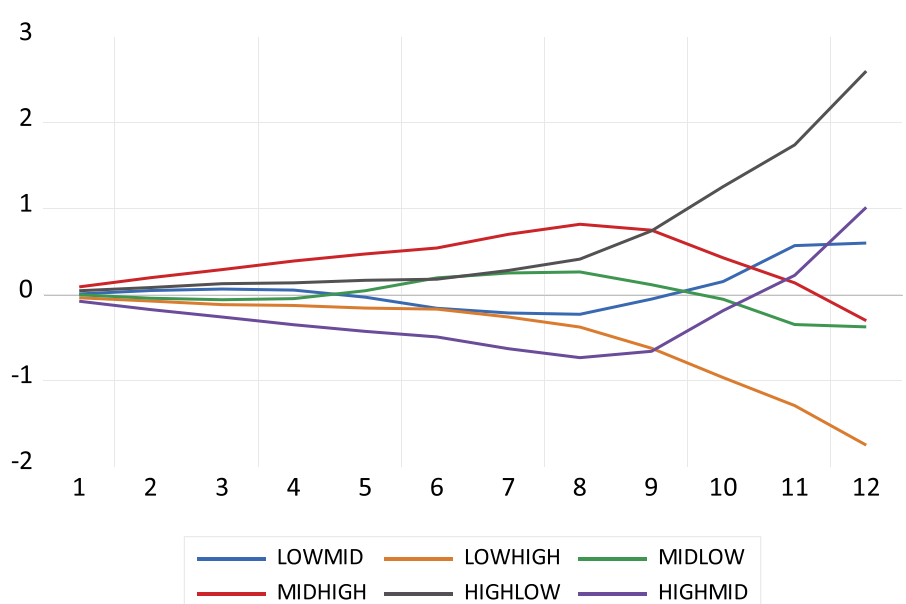
4.3. Spillover indices across portfolios

Panels A and B of Table 6 report the estimates of the spillover indices based on 10-year ahead forecast error variance decompositions. The decompositions are on residuals from VAR models, including estimated risk-weighted standard deviations on low-, mid-, and high-income countries denoted by $\delta_{low,t}$, $\delta_{mid,t}$, and $\delta_{high,t}$, respectively.

Table 6 also includes Panel B reporting the diagonal entries from 12 VAR models, where the first VAR is of order 1 and the last VAR is of order 12. We are aware that with only 56 yearly observations, higher VAR models may be over-identified. However, the purpose of this exercise is to examine whether volatility spillovers increase with years. Clearly, a decreasing own variance indicates that the VAR system improves and the contributions "to others" and "from others" increase in proportion.

The total volatility spillovers in the lower right corner of Panel A and B are 53.50% and 65.60% on VAR (1) and VAR (12), respectively. The total volatility spillover index provides a general appreciation of volatility transmission across countries. An increase of 22.62% over 12 years suggests that the interlinkage gets stronger with years. However, we would like to examine which group of countries is responsible for such a long delay in volatility transmission. Panel C reports changes in the diagonal entries of the 12 matrices for each group of countries, showing that changes in own variance of high-income countries turn positive in year 2, which is a first indication that spillovers from the other two groups to the high-income growth group may have fully happened in year 1. Panel C of Table 6 uses changes over the diagonal entries for each group of countries to examine whether the own variance of low-, mid-, and high-income countries decline over time.

We test the average changes at lag 6. The averages (the standard errors) of the low-, mid-, and high-income growth group are −0.8% (0.32%), −0.41% (0.30), and −0.46 (0.34%), respectively. Based on the Z-scores, the low-income growth group is the slowest at absorbing "global" volatility shocks. Repeating the test at lag 12 with 11 changes included, the high-income growth group remains immune from volatility



Note: Figure 4 is constructed on pair-wise net volatility entries of Table 6.

Fig. 4. Cumulative net volatility transfer patterns over the years.

spillovers that may originate from either low- or mid-income growth group.

Since Table 6 shows that it would take several years for global shocks to be absorbed by most of the countries in the low-income growth group, it is worth examining the dynamics of the net effect of transmitting to and receiving from others. Since the low-income growth group is slower at incorporating global volatility shocks, countries in this group may not be contributing significantly to the overall global volatility.

Looking at Panels A and B of Table 6 from the perspective of a country in the low-income growth group, pairwise net volatility transfer performances are given by $\phi_{low,mid} = (a_{21} - a_{12})/a_{12}$ and $\phi_{low,high} = (a_{31} - a_{13})/a_{13}$ with respect to a country in the mid- and high-income growth group, respectively. Table 7 reports pairwise net volatility spillover for each lag in the VAR system for the six possible combinations.

A negative entry means the group is a net volatility receiver, while a positive entry means the group is a net volatility giver. The entries are not necessarily symmetric in magnitude. For instance, at lag 1, low-income countries are receivers with respect to high-income countries, while mid-income countries are transmitters with respect to high-income countries. Table 7 also shows that the transfer from the low-income countries to the high-income countries is negative at all lags, which is an indication that the growth volatility dynamics of low-income countries are markedly disconnected from the growth volatility dynamics of high-income countries.

Testing whether the transfer of volatility among groups is significant, we found that over 6 and 12 years, low-income countries are significant net receivers of volatility from high-income countries. In other words, high-income countries are immune to volatility effects from low-income countries.

While a significant transfer of volatility between low- and mid-income countries seems inexistent, the latter are significant net volatility transmitters to high-income countries. However, unlike the transmission of volatility from high-to low-income countries that seems to be occurring for years, the transmission between mid- and high-income countries is not significant at year 12.

Given that we found that entries are normally distributed, the following two main conclusions emerge from Table 7. First, high- and low-income countries move in opposite directions, whereas mid- and low-income countries do not significantly share volatility spillovers over the years. Second, mid-income countries are stronger volatility transmitters to high-income countries than high-income countries are to low-income countries. Fig. 4 portrays the cumulative effects of net volatility transfers among group of countries.

Fig. 4 displays two interesting volatility paths. First, the volatility path portraying mid- and high-income countries. Right at the root of the two volatility paths, there is a daylight difference between MIDHIGH and HIGHMID, as the two curves are the envelopes. While the two curves diverge, they converge at lag 11, with the curve referred to as HIGHMID turning positive. Second, the curves referred to as LOWHIGH and HIGHLOW (giving the perspective of low- and high-income countries, respectively) diverge right from the start.

Fig. 4 may possibly capture a number of economic phenomena such as volatility divergence and economic disconnection in a time of economic crises. Fig. 4 also shows some peculiarity since the spread between the two curves widens from year 5 onwards. There is a high possibility that these two groups of countries represent external envelopes of the world economy. However, looking at GDP growth alone is not enough to draw definite conclusions on how disconnected low-income countries are from high-income countries. We therefore reexamine our VAR system of three equations in terms of macroeconomic volatility[10].

Table 8
Macroeconomic volatility interlinkage indices.

Panel A: Volatility spillover weights on growth and macroeconomic volatility variables for countries in the low-income group

	W_{gw}	W_{mx}	W_{bm}	W_{cr}	W_{fi}	W_{FO}
W_{gw}	0.6036	0.1373	0.1290	0.0665	0.0636	0.3960
W_{mx}	0.1075	0.4370	0.3267	0.0039	0.1250	0.5630
W_{bm}	0.1082	0.3287	0.4097	0.0066	0.1469	0.5900
W_{cr}	0.0824	0.0102	0.0230	0.7462	0.1382	0.2540
W_{fi}	0.1119	0.0400	0.0231	0.0025	0.8224	0.1780
W_{TO}	0.4100	0.5160	0.5020	0.0800	0.4740	1.9810
$W_{ii} + W_{TO}$	1.0140	0.9530	0.9120	0.8260	1.2960	0.3960

Panel B: Volatility spillover weights on growth and macroeconomic volatility variables for countries in the mid-income group

	W_{gw}	W_{mx}	W_{bm}	W_{cr}	W_{fi}	W_{FO}
W_{gw}	0.7925	0.0846	0.0422	0.0785	0.0023	0.2070
W_{mx}	0.0689	0.4838	0.4269	0.0198	0.0006	0.5160
W_{bm}	0.0303	0.4569	0.5101	0.0008	0.0018	0.4900
W_{cr}	0.0726	0.0627	0.0477	0.8119	0.0052	0.1880
W_{fi}	0.0291	0.0607	0.0225	0.0596	0.8280	0.1720
W_{TO}	0.2010	0.6650	0.5390	0.1590	0.0100	1.5740
$W_{ii} + W_{TO}$	0.9930	1.1490	1.0490	0.9710	0.8380	0.3150

Panel C: Volatility spillover weights on growth and macroeconomic volatility variables for countries in the high-income group

	W_{gw}	W_{mx}	W_{bm}	W_{cr}	W_{fi}	W_{FO}
W_{gw}	0.7553	0.0252	0.0196	0.0788	0.1210	0.2450
W_{mx}	0.0121	0.3878	0.2003	0.0050	0.3947	0.6120
W_{bm}	0.0195	0.2263	0.4471	0.0132	0.2939	0.5530
W_{cr}	0.0931	0.0200	0.0417	0.8072	0.0380	0.1930
W_{fi}	0.0035	0.0216	0.0292	0.0022	0.9436	0.0560
W_{TO}	0.1280	0.2930	0.2910	0.0990	0.8480	1.6590
$W_{ii} + W_{TO}$	0.8840	0.6810	0.7380	0.9060	1.7910	0.3320

Notes: Spillover indices are based VAR(1) models and variance decompositions based on 10-step-ahead forecasts. The VAR model includes σ_{gw}, σ_{mx}, σ_{gw}, σ_{bm}, σ_{cr}, and σ_{fi}, where σ_{ii} is standard deviation with ii = growth (gw), imports/exports (mx), broad money (bm), credit to the private sector (cr) and net foreign direct investment (fi). W_{ii}, W_{TO}, and W_{FO} are diagonal entries for each ii, the sum of weights "to others" and "from others," respectively.

4.4. Growth and macroeconomic volatility dynamics

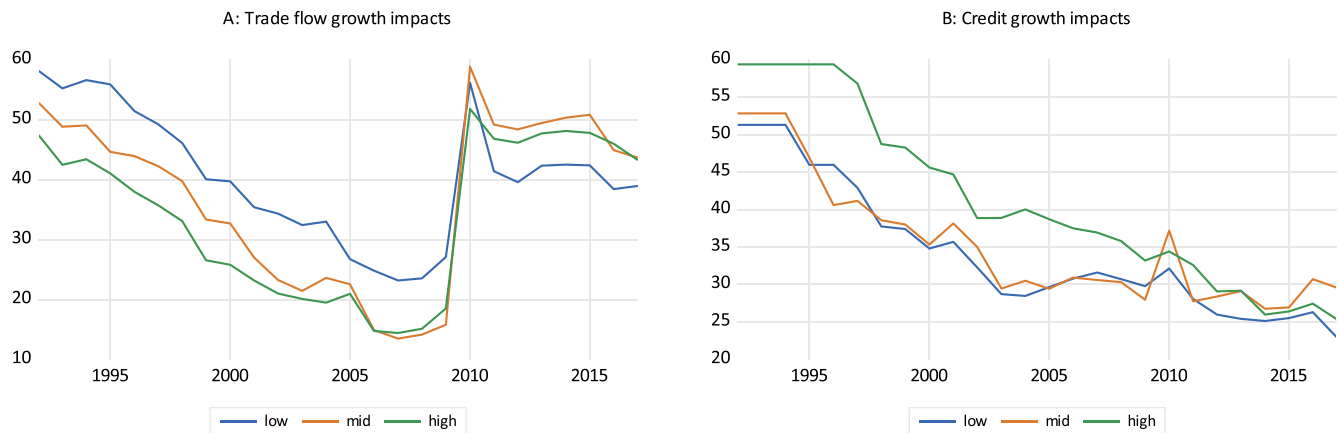
Let $\delta_{p,gw,t}$, $\delta_{p,mx,t}$, $\delta_{p,bm,t}$, $\delta_{p,cr,t}$, and $\delta_{p,fi,t}$ be the risk-weighted standard deviations of growth, imports plus exports, broad money and FDI, respectively. We estimate for group p, using sample B, a VAR model of order 1, where the order is determined by the Akaike Information Criterion. Table 8 reports the estimates of the volatility spillover indices based on 10-year ahead forecast error variance decompositions.

Panels A, B and C show that growth is a poor transmitter of shocks to macroeconomic volatility for mid and high-income countries. In contrast, for low-income countries, shocks in growth may lead to higher macroeconomic volatility looking only from the perspective of either "from others" or "to others."

Panels A, B and C also show that FDI does not greatly contribute to the total overall volatility because its diagonal entries range from 82.24% to 94.36%. Nonetheless, the FDI contribution to growth is 12.10% for high-income growth countries. FDI is also an important transmitter of shocks to MX, BM, and CR for low-income countries.

Since Table 8 seems to indicate that volatility dynamics are stronger among low-income countries, the ratio of "from others" to "to others" or

[10] We also did in parallel a robustness check on how results on ranking countries in terms of the ratio of growth to growth standard deviation (CUR) differ from results on ranking in terms of GEF, RUL, and REQ. We find that GEF, RUL, and REQ rank industrialized countries except for China, quite well among the high-income countries, but rank poorly mid- and low-income countries. As we also looked at the variance dynamics in volatility transfers, we find that GEF, RUL, and REQ ranking give unintuitive volatility transfer patterns. Comparing for instance Fig. 4 that is based on the CUR ranking with graphs on alternative ranking, we find that CUR is better at describing growth volatility dynamics over time. The results on the basis of GEF, RUL, and REQ ranking are available upon request.



Notes: Spillover indices are based on VAR(1) models and variance decompositions based on 10-step-ahead forecasts. The three VAR models of Panel A includes $\sigma_{p,gw}$, and $\sigma_{p,mx}$, where $\sigma_{p,gw}$ is the standard deviation of GDP growth on portfolio p, and $\sigma_{p,mx}$ is the standard deviation of imports plus exports (mx) on portfolio p. For instance, the VAR equations for the low-income growth group include $\sigma_{low,gw}$, $\sigma_{low,mx}$, $\sigma_{mid,mx}$, and $\sigma_{high,mx}$. The three VAR models of Panel B includes $\sigma_{p,gw}$, and $\sigma_{p,cr}$, where $\sigma_{p,gw}$ is the standard deviation of GDP growth on portfolio p, and $\sigma_{p,cr}$ is the standard deviation of credit flow (cr) on portfolio p. The vertical axis gives total spillover indices obtained by a rolling-window of 30 years.

Fig. 5. Growth, trade flow, and credit flow volatility dynamics.

(W_{FO}/W_{TO}) should highlight factor with critical impact for each group. Let us consider the case where $(W_{FO}/W_{TO}) > 1$, which refers to higher vulnerability. Then BM and CR are critical factors for low-income countries, while GW, MX, BM, and CR are critical for high-income countries.

Fig. 5 summarizes the rich information in Table 8, building on VAR (1) models including growth and trade volatility, and growth and credit flow volatility in Panels A and B, respectively.[11] The vertical axes give total volatility spillover indices obtained on 30-years rolling windows. It follows that the higher the total volatility index, the greater the spillover.

Fig. 5 captures, as in Awartani and Maghyereh (2013) on daily data, Antonakakis and Badinger (2016) on monthly output data, and Kang et al. (2017) on weekly equity data, the sharp increase of growth and trade volatility dynamics in 2009. However, this shift in volatility of 2009 does not appear in growth and credit flow volatility dynamics, which is an indication of a partial economic and financial recovery in the world after 2009.

Fig. 5 also shows that, prior to the global financial crisis of 2008/2009, growth and trade volatility dynamics were much stronger for low-income countries than for high-income countries. Following the financial crisis, mid-income countries have contributed the most to volatility spillovers.

Contrasting Panel A with Panel B, trade turns out to be an important source of growth for low-income countries, while credit to the private sector is a more important (but depleting) source of growth for high-income countries.

5. Summary and concluding remarks

We take a portfolio perspective to investigate growth volatility dynamics among various economies. The portfolio approach allows us to include virtually all the countries of the world, and rank them in terms of their "empirical performance," while mitigating the endogeneity problem particularly acute for countries sharing the same information structure. Our analysis of volatility dynamics provides spillover proportions, which are important inputs in portfolio strategies, hedging activities, and risk management. Instead of deriving spillover proportions from MGARCH models, we rely on VAR models under Diebold and Yilmaz (2012), which are enhanced with impulse response functions at different forecasting horizons.

Our estimates of own-volatility proportion patterns indicate that it takes several years for low-income countries to absorb volatility effects from global origins. The bidirectional spillovers show that low-income countries are highly disconnected from the rest of the world. Since low-income countries may contribute to common volatility through different channels, we examine how macroeconomic volatility relates to growth volatility, and find that trade is an important source of growth volatility. We also find that while credit to the private sector is a key factor of growth volatility, its contribution to volatility has been considerably declining over time.

Overall, our results on volatility dynamics suggest that investment strategists can capitalize on our volatility proportions in detecting countries with greater growth potential. Countries' idiosyncratic contributions to common volatility can help identify countries with long-term investment opportunities associated with higher growth volatility. Since specific macroeconomic conditions prevail in low-income countries, volatility risk must be fully diversified internally.

Acknowledgement

We would like to thank the Editor (Dr. Sushanta Mallick) and the two referees for their helpful comments. We also thank participants at the AKSOB seminar series for their comments.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.econmod.2019.11.015.

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